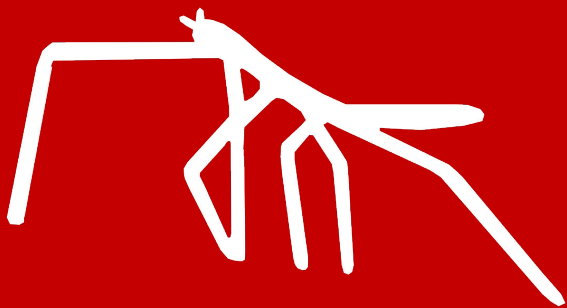


Boing Limited



STICK BUG

Final Design Review

Athanasios Stefanos, Bradley Spiers, Dion Castelino, Jesse Laity,
Khuong Nguyen, Matthew Hayes, Patrick Dragovic, Pratham Patel,
Theo Fry, Muhammad Zain Ul Abidin



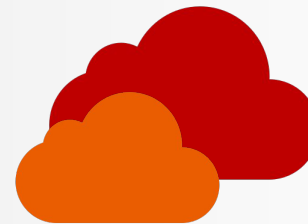
DESIGN REQUIREMENTS



250 PAX
+ 8 Crew



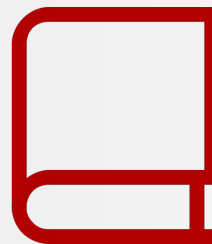
7500 - 8000 km
Range



35,000 ft
Service Ceiling



0.85
Target Mach



CASR P.25
Certification

MARKET OPPORTUNITY



Air traffic expected to overtake
pre-COVID levels 2 years

\$1.2B New York → London
market suited to 250 PAX

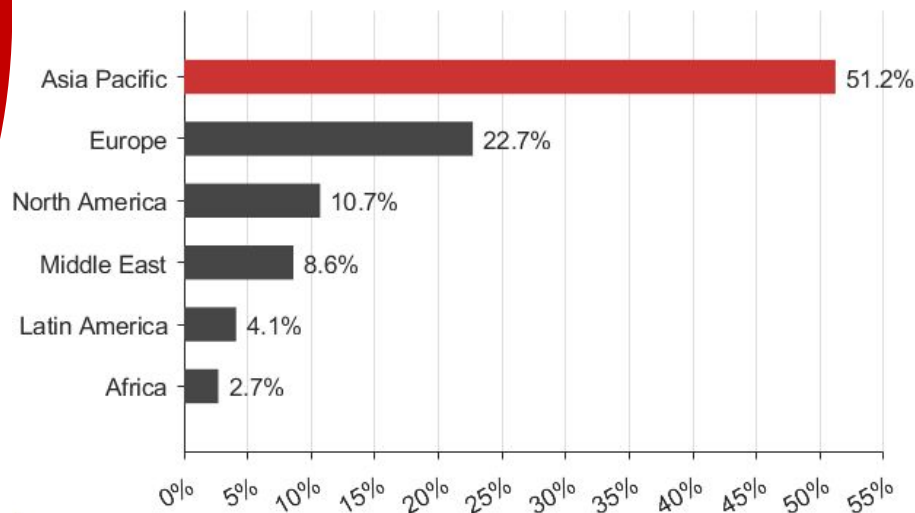


15% lower DOC allows \$20-30
USD decrease in airfares

Asia-Pacific region offers
fastest growth annually



GLOBAL PASSENGER TRAFFIC GROWTH +



COMPETITIVE LANDSCAPE

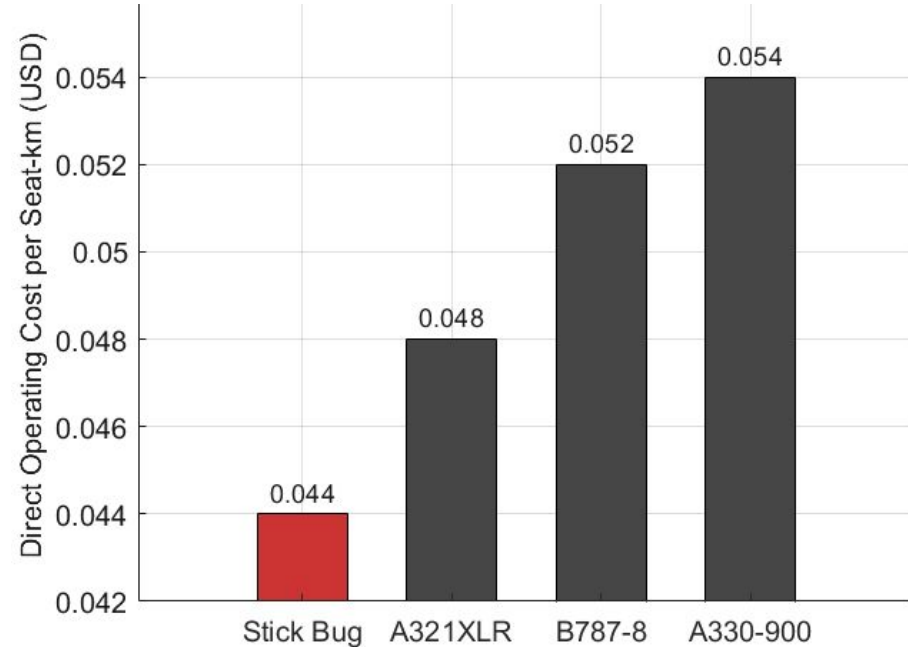


%15 DOC margin on competition planes



Trent 1000-E offers low fuel burn and reduced noise pollution

ICAO Type-E compliance to enable airport accessibility



Stick Bug records an average operating cost of **0.044 USD per seat-km**, outperforming comparable long-range competitors.

Overview



Cabin Configuration

28 Business Class Seats
222 Economy Seats
Double Aisle, 7 Abreast Economy Seating
Double Aisle, 4 Abreast Business Class Seating
7 Lavatories, 3 Galleys
4 Pairs of Type-A Emergency Exits

Capabilities

High Speed Harmonic Range of 7434 km at Mach 0.85
Long Range Harmonic Range of 7676 km at Mach 0.80
24.18 Lift to Drag Ratio at Cruise
45000 ft Service Ceiling

Logistics

8 Crew Members
44 LD3 Cargo Containers

Propulsion

2 Rolls-Royce Trent 1000-E Engines
530 kN Maximum Thrust
0.4458 Aircraft Thrust to Weight Ratio

Landing Gear

Tricycle Landing Gear
Michelin Air X Tyres
23.15 m Wheelbase
13.09° Tail Strike Angle

Aerodynamic Surfaces

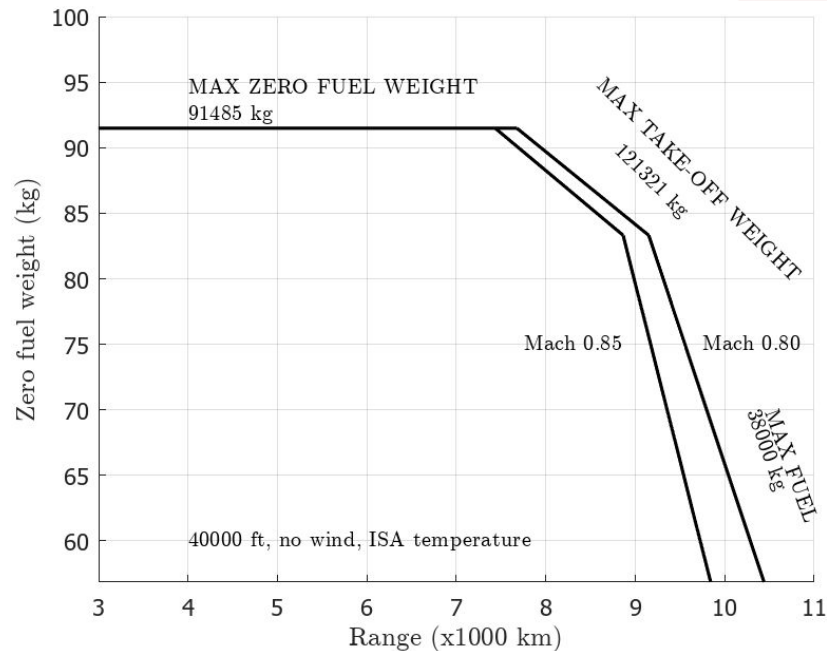
Strut Braced Wing
58.17 m Wingspan, 15.29 Aspect Ratio
T-Tail Horizontal Stabiliser Configuration
Slats and Double Slotted Flaps





PAYLOAD – RANGE

RANGE	HARMONIC	MAXIMUM
Long-range cruise (M0.80)	7676 km (4143 nm)	9149 km (4937 nm)
High-speed cruise (M0.85)	7434 km (4012 nm)	8863 km (4738 nm)



PERFORMANCE

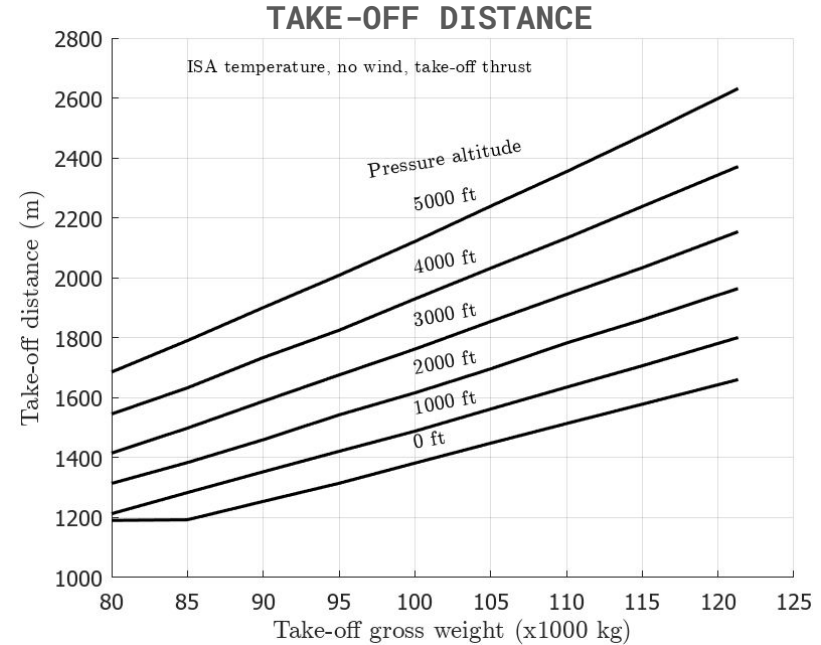


- Iterative integration analysis of lift, drag, thrust and angle of attack

Take-off

MTOW, SSL conditions

- Take-off run: 1421 m (4665 ft)
- Take-off distance: 1660 m (5446 ft)
 - V_{MU} : 127 KTAS
 - V_R : 133 KTAS
 - V_{LOF} : 139 KTAS
 - V_2 : 143 KTAS



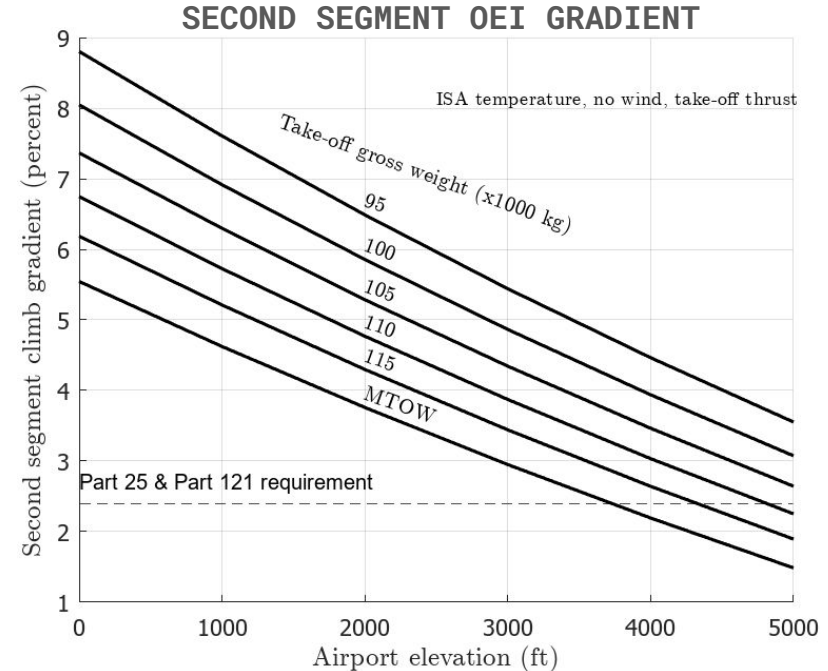
PERFORMANCE

- Iterative integration analysis of lift, drag, thrust and angle of attack

Take-off - second-segment gradient

2.4 % one-engine inoperative requirement

- 5.7 % at MTOW, SSL
- Satisfied for airport up to **3600 ft** elevation at MTOW



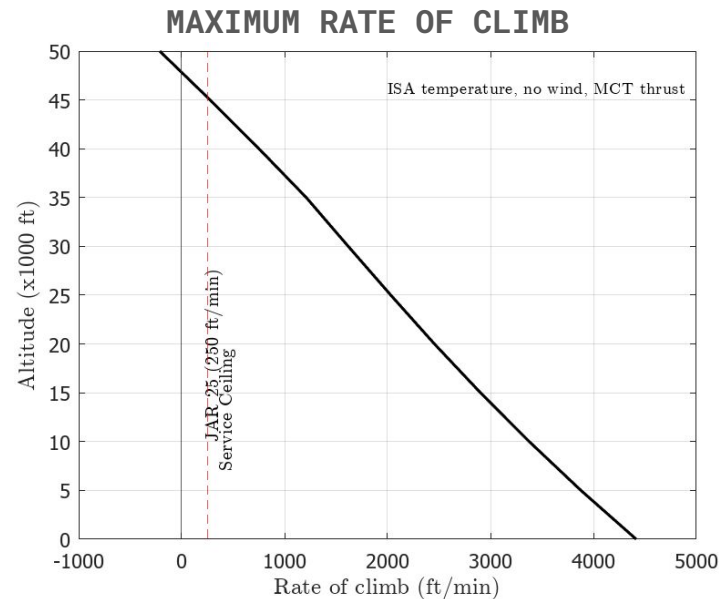
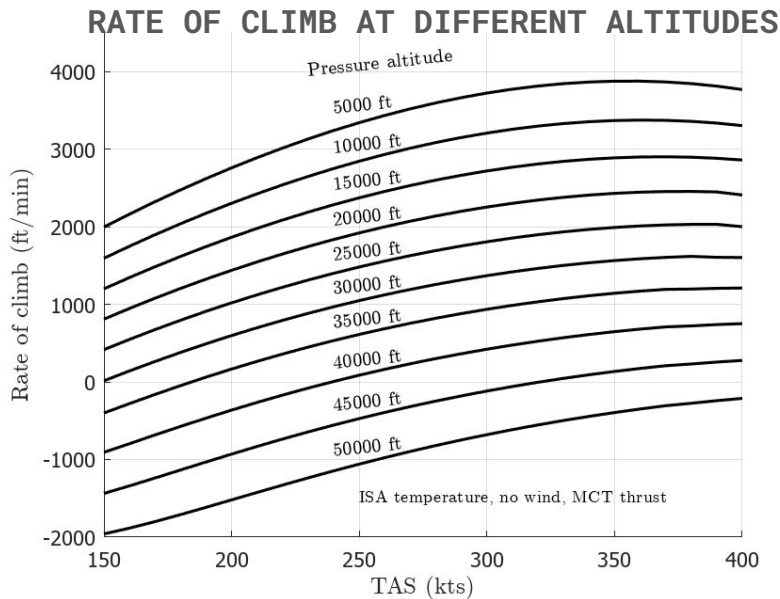
PERFORMANCE



En-route climb

Climb performance evaluated at different pressure altitudes for MTOW

- Max rate of climb (SSL): 4320 ft/min
- Service ceiling: 45,000 ft



PROPULSION SYSTEM



Engines	Rolls-Royce Trent 1000-E
Bypass Ratio	11:1
Thrust	265.3 kN (each)

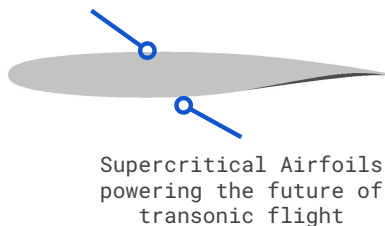
Engine Upgrade	Rolls-Royce Ultrafan
Fuel Improvement	25% on previous engines + SAF
Timeline	2030's
Opportunity	Hydrogen / hybrid-electric

UNMATCHED PERFORMANCE

WING OPTIMISATION

Aspect Ratio (AR)	15.3
Sweep	26.5 deg
Wingspan	64 m
Root Chord	6.68 m
Tip Chord	1.72m

Optimised Elliptical Design,
Minimal drag - Maximum performance



First of its kind 15 AR wings
Increased Aerodynamic Efficiency
Increased Fuel Savings



TAKE OFF & LANDING

Maximising Performance

**Takeoff
Distance**

1660 m

Landing Speed

67.2 m/s

Maximum AoA

12 degrees

Reliability

Rack and Pinion
Drive

Four Bar Linkage

Operational

Maximal
Performance at
cruise

Maximal
performance on
the runway



TAIL LAYOUT

Supercritical airfoils to avoid shocks forming at cruise

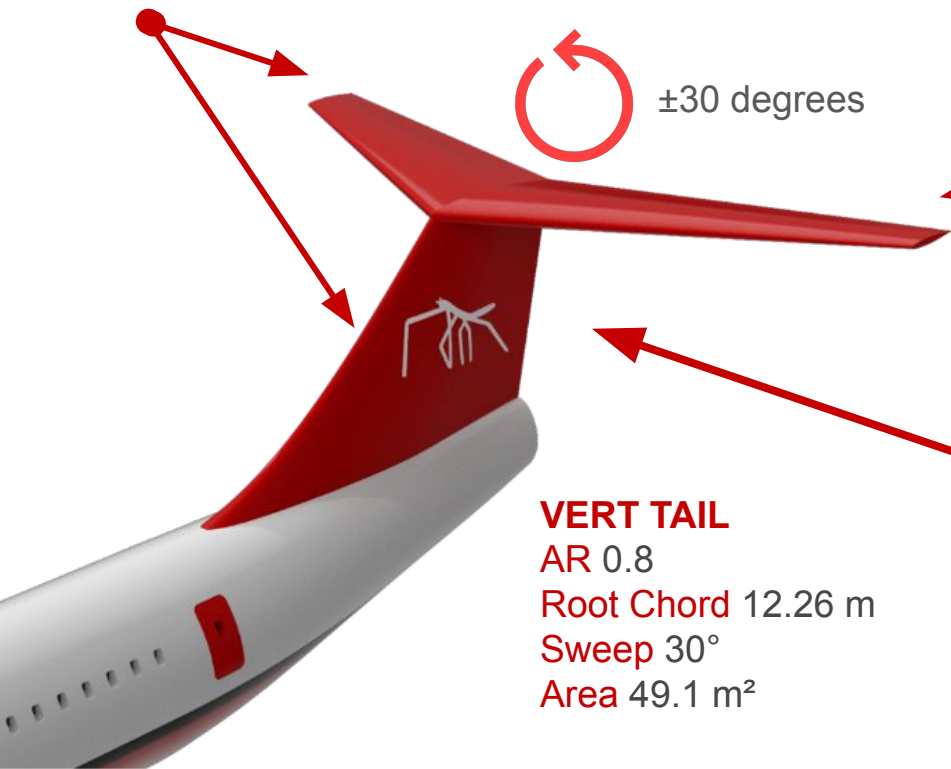
ELEVATOR

AR 6.5

Root Chord 5.05 m

Sweep 26.5°

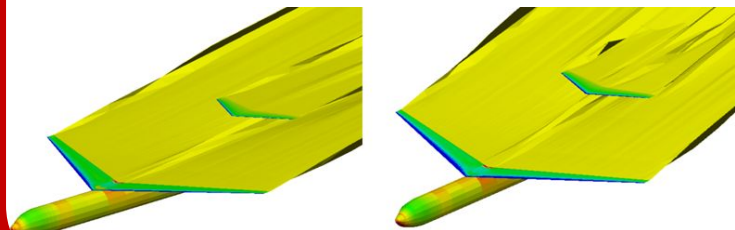
Area 81.1 m^2



± 30 degrees

All-moving elevator for trim and stability

T-Tail configuration to avoid blanketing at high AoA conditions (Landing & takeoff)



VERT TAIL

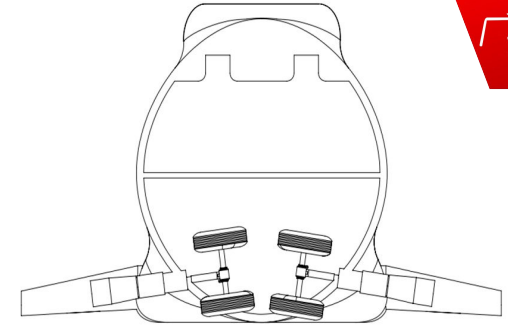
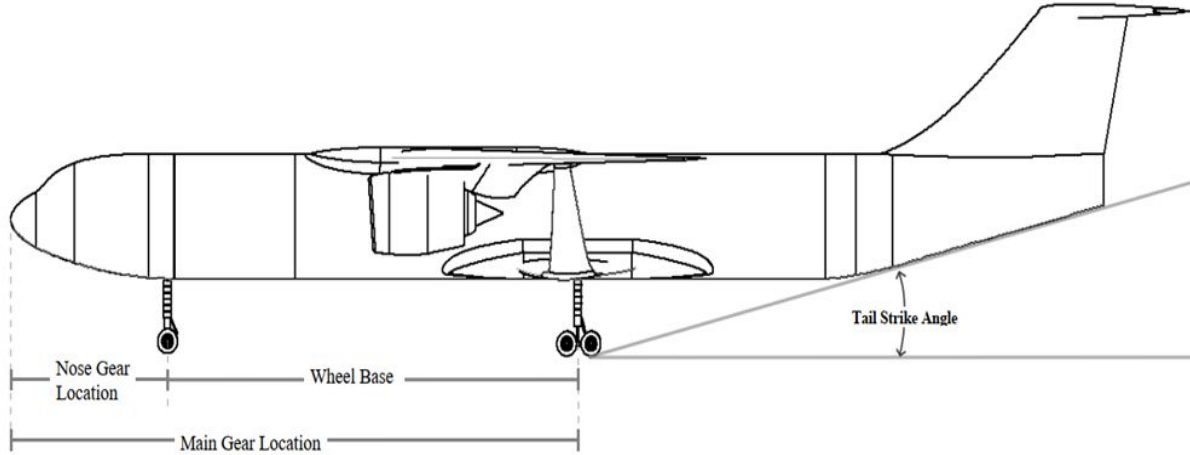
AR 0.8

Root Chord 12.26 m

Sweep 30°

Area 49.1 m^2

Landing Gear

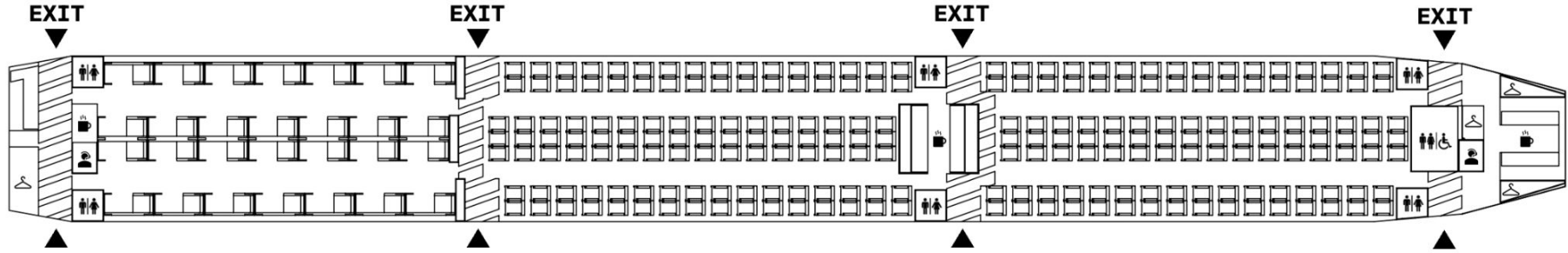


Features

Parameter	Value	Parameter	Value
Tip Back angle	20.76°	Main gear from nose	32.0m
Tail strike angle	13.09°	Nose gear from nose	8.85m
Wheel track	6.75m	Undercarriage height	3.58m

- Michelin Air X Tyres
- Telescopic retraction mechanism for adequate stowage
- Tricycle configuration for superior ground handling and stability

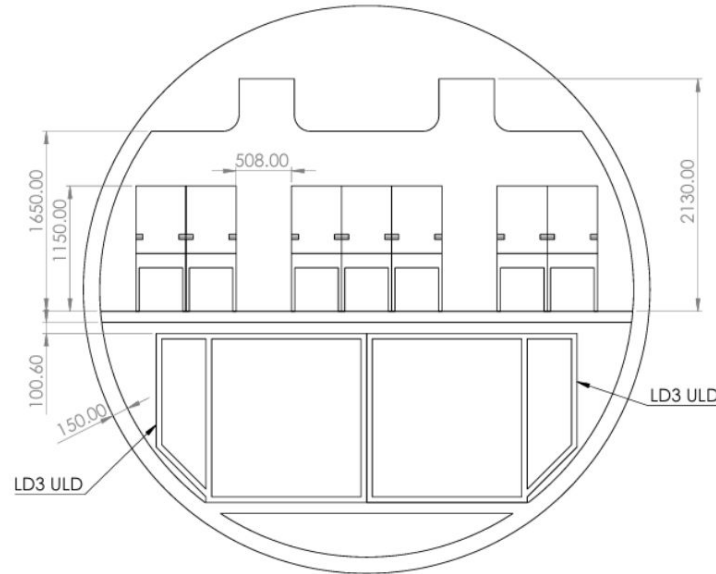
CABIN LAYOUT



28 Business
Class

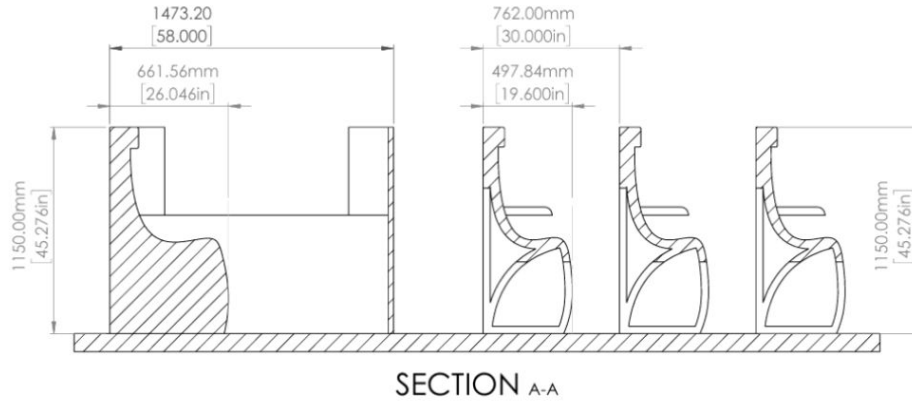
222 Economy
Seats

48 LD3 ULD
Storage



Excess bulk Storage:
Opportunity for courier
cargo

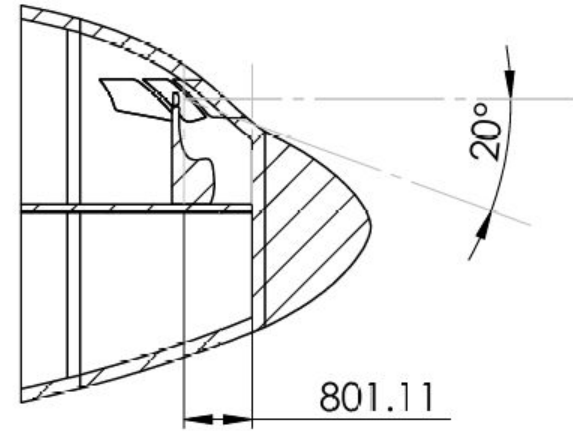
CABIN LAYOUT



E-class seat width is 460 mm (18.1"), and 762 mm (30 inch) in pitch

B-class seat width is 533 mm (21") and a spacious 1473.20 mm (58") - inspired by the JAL Sky Suite.

Aisle width is 20" in compliance with FAR 25.815 for seats height 25" above the ground.



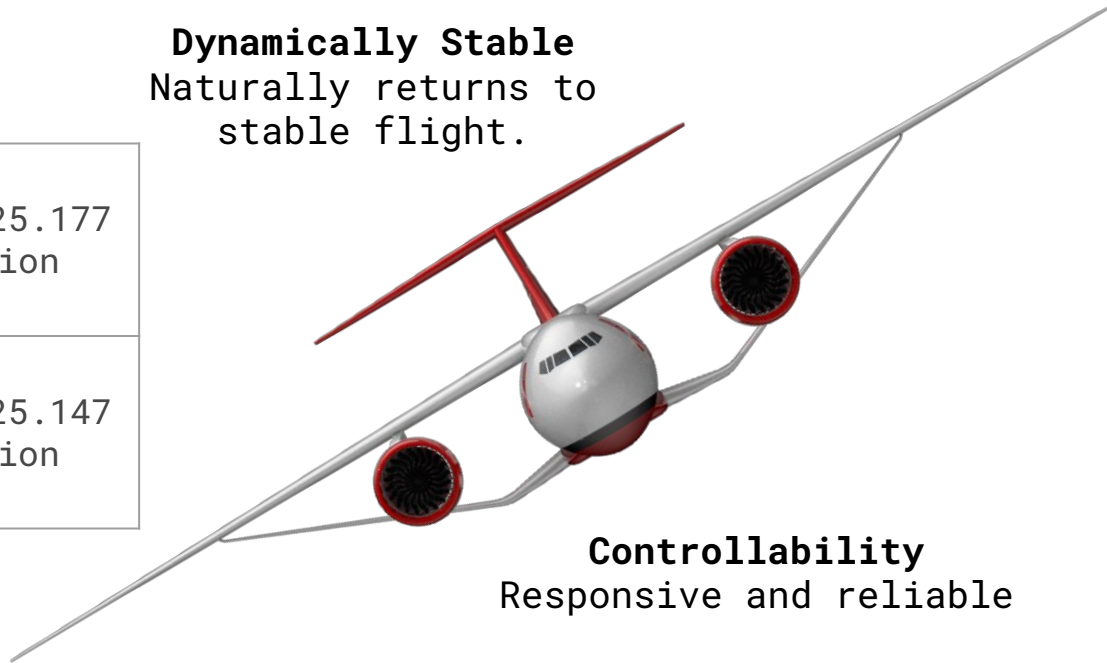
Pilot is position with the 20° of downward view, as suggested by Raymer (18-20°), with the V-shaped window design extending behind the pilot for > 180° field of view.

Pilot Handling and Control



Static Stability	Exceeds CFR 25.177 Certification
Lateral Stability	Exceeds CFR 25.147 Certification

Dynamically Stable
Naturally returns to stable flight.

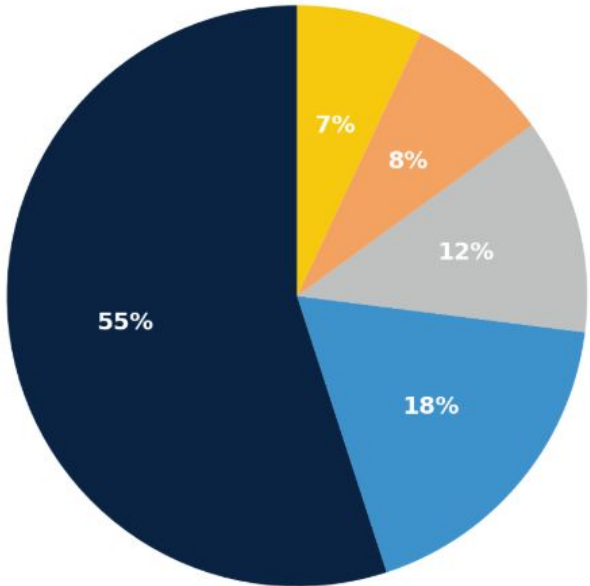


Controllability
Responsive and reliable

Predictable Handling
Smooth response



COST & PRICE ANALYSIS



Direct Operating Cost — \$70 M USD per year

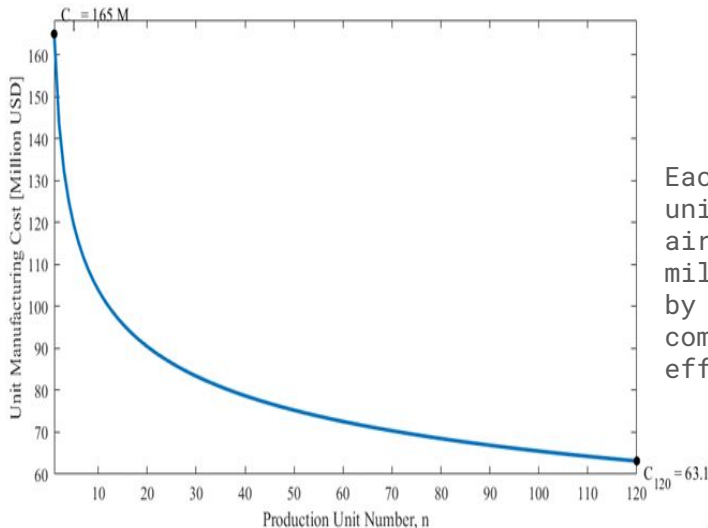
- Fuel — \$38.5M
- Maintenance (Airframe + Engine) — \$12.6M
- Depreciation & Insurance — \$8.4M
- Miscellaneous / Ground Ops — \$5.6M
- Crew & Training — \$4.9M

Aircraft Economics:

- List price = US \$ 238 million
- Cost per seat $\approx$ US \$ 0.95 million (approx 25% below comparable aircrafts)
- Operating cost per seat-km $\approx$ 15% lower than boeing 787-7 / A330-300

Financial performance:

- Payback period $\approx$ 12 years
- Net present value = US \$ 84 million per aircraft
- ROI $\approx$ 17%, improving with fleet scale



Each production doubling reduces unit cost by about 13%. The first aircraft costs around US \$165 million, dropping to US \$63 million by the 120th unit as automation and composite manufacturing become more efficient.

Questions?

Boing Limited

